

1. Write the equation of the line that passes through (4,2) and is parallel to $y = 3x + 1$ in point-slope form.

$$y - 2 = 3(x - 4)$$

2. Write the equation of the line that is parallel to $y = -\frac{3}{2}$ and passes through point (2,3).

$$y = 3$$

3. Write the equation of the line that contains point (0,5) and is parallel to $2x + 3y = 6$ in slope-intercept form.

$$2x + 3y = 6 - 2x$$

$$5 = -\frac{2}{3}(0) + b$$

$$-2x$$

$$5 = b$$

$$\frac{3y}{3} = \frac{-2x+6}{3}$$

$$y = -\frac{2}{3}x + 2$$

$$y = -\frac{2}{3}x + 5$$

4. Write the equation of the line that is parallel to $x - 2 = 0$ and passes through point (1,-2).

$$x = 1$$

5. Write the equation of the line that passes through the origin and is parallel to $y + 9 = -(x - 2)$ in standard form.

$$y - 0 = -1(x - 0)$$

$$y = -x$$

$$x + y = 0$$

6. Write the equation of the line, in all three forms, that passes through point $(-10, -3)$ and is parallel to $y + \frac{1}{4} = \frac{1}{4}(x - 1)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - (-3)$ $= \frac{1}{4}(x - (-10))$ $y + 3 = \frac{1}{4}(x + 10)$	$-3 = (-10)\left(\frac{1}{4}\right) + b$ $-3 = -\frac{10}{4} + b$ $+\frac{10}{4} \quad +\frac{10}{4}$ $-\frac{1}{2} = b$ $y = \frac{1}{4}x - \frac{1}{2}$	$(y + 3 = \frac{1}{4}x + \frac{10}{4}) \cdot 4$ $4y + 12 = x + 10$ $-x - 12 \quad -x - 12$ $-x + 4y = -2$ or $x - 4y = 2$

7. Write the equation of a line that is parallel to $y = x$ and contains point $(-2, 2)$.

$$2 = 1(-2) + b$$

$$2 = -2 + b$$

$$+2 + 2$$

$$4 = b$$

$$y = x + 4$$

8. Line N passes through $(\frac{1}{2}, -6)$ and is parallel Line Q which has an undefined slope. Write the equation of Line N .

$$x = \frac{1}{2}$$

9. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(0, 5)$ and $(\frac{5}{6}, 0)$ and passes through point $(4, 4)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{0-5}{\frac{5}{6}-0} = \frac{-5}{\frac{5}{6}} = -6$ $y - 4 = -6(x - 4)$	$4 = 4(-6) + b$ $4 = -24 + b$ $+24 + 24$ $28 = b$ $y = -6x + 28$	$y - 4 = -6x + 24$ $+6x + 4 + 6x + 4$ $6x + y = 28$

10. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(-6, 1)$ and $(10, 0)$ and passes through point $(-7, -2)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{0-1}{10-(-6)} = \frac{-1}{16}$ $y - (-2) = \frac{-1}{16}(x - (-7))$ $y + 2 = \frac{-1}{16}(x + 7)$	$-2 = (-7)\left(-\frac{1}{16}\right) + b$ $-2 = \frac{7}{16} + b$ $\frac{-7}{16} \quad \frac{-7}{16}$ $-\frac{39}{16} = b$ $y = \frac{-1}{16}x - \frac{39}{16}$	$y + 2 = \frac{-1}{16}x - \frac{7}{16} - 2$ -2 $\left(y = \frac{-1}{16x} - \frac{39}{16}\right) 16$ $16y = -x - 39$ $+x + x$ $x + 16y = -39$